

# How HMI Technologies Are Enhancing Safety And In-Vehicle Experience

As we are moving towards smart, connected, and autonomous mobility, the interaction between the vehicle and the human has also evolved. The HMI solutions of today not only enhance the in-vehicle experience and convenience but also deliver personalised experiences. Only the information that is critical at a particular moment is communicated to the driver

Rosemary Joshy

**V**ehicles are transforming from a means of transportation to mobile living spaces. The future of mobility is driven through innovations that make the vehicles safer, seamlessly connected to the mobile ecosystems, and more convenient for drivers and passengers. Safe, connected, convenient are the critical parameters that the consumers today are evaluating for their buying decision. With autonomous and

smart mobility not any longer a discussion for tomorrow, rather a reality of today, the drivers will be co-drivers or passengers soon. Hence one of the major differentiators is the user experience that we can provide.

Human machine interface (HMI) provides this great user experience to see, feel, and hear through the design excellence, technology, and intelligent information management. It has made in-vehicle experiences

more connected, informed, and integrated. Further, according to the Future Automotive Industry Structure—FAST 2030 report, HMI is one of the seven trends that will reshape the automotive industry by 2030.

## HMI technologies that will enhance safety and in-vehicle experience

Long ago the radio, warning chimes, and, of course, the passengers were the sounds we could hear in the vehicle, but now the vehicle itself is becoming a smart communication partner. Vehicles have more to talk than ever before. An ever-increasing number of sensors and the vehicle being a part of the Internet of Everything create



*Glasses-free 3D effect: The Lightfield Display allows a comfortable perception of 3D depth as well the rendering of highlights and other light effects*